

Fig. 3: Actual-size PCB pattern of the transmitter unit

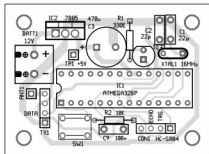


Fig. 4: Component layout of the PCB shown in Fig. 3

PARTS LIST

Semiconductors:

- IC1, IC3 - ATmega328P-PU MCU with Arduino Uno bootloader
- IC2, IC4 - 7805, voltage regulator
- TX1 - 433MHz RF transmitter module
- RX1 - 433MHz RF receiver module
- LED1, LED2 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3 - 330-ohm
- R2, R4 - 10-kilo-ohm
- VR1 - 10-kilo-ohm potentiometer

Capacitors:

- C1, C2, C4, C5 - 22pF, ceramic disk
- C3, C8 - 470 μ F, 25V electrolytic
- C6 - 10nF, ceramic disk
- C7 - 10 μ F, 16V electrolytic
- C9, C10 - 10nF, ceramic disk

Miscellaneous:

- BATT1, BATT2 - 12V battery
- ANT1, ANT2 - Spiral antenna
- XTAL1, XTAL2 - 16MHz crystal oscillator
- LCD1 - 4x16 alphanumeric display
- SW1, SW2 - Tactile switch
- CON1 - 4-pin connector for HC-SR04 ultrasonic ranging module

provides 20mm to 5000mm non-contact measurement facility. Ranging accuracy is 3mm and aperture angle is 15° . Since we are measuring the level in centimetres in the form of an integer, our accuracy level is maximum 5mm. Even a 1cm level change will be seen very clearly, as every second it takes eight readings,

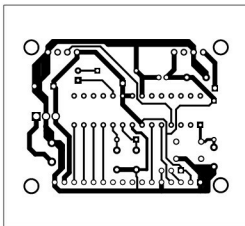


Fig. 5: Actual-size PCB pattern of the receiver unit

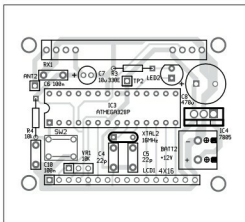


Fig. 6: Component layout of the PCB shown in Fig. 5

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

which are then averaged out to get better accuracy.

Circuit diagram of the receiver unit is shown in Fig. 2. It is built around ATmega328P MCU (IC3) with Arduino Uno bootloader, voltage regulator 7805 (IC4), 4x16 alphanumeric display (LCD1), 433MHz receiver (RX1) and a few other components. The receiver checks the code word sent by the transmitter unit and displays the tank level on the LCD.

If the transmitter stalls or its power supply gets interrupted, there is no way for the receiver to find out that the incoming signal is no longer

valid. To circumvent this problem, a counter has been provided on the right side of the LCD display. If the counter does not move, or it stops, it means that the incoming signal has stalled. Blinking of LED2 connected on pin 19 will stop, too.

Control relay and contactors can be connected on spare pins of ATmega328P, which finally controls the running of the water pump. Since there is no physical contact used in the sensory system, it will be operational without any interruption.

Software

Software comprises one remote_sonar_water_level_transmitter.ino file and one remote_sonar_water_level_receiver.ino file, besides some header files of Arduino. All are packed in the resource directory. Add these libraries to Arduino and upload the two files in the respective transmitter and receiver units.

Construction and testing

An actual-size, single-side PCB of the transmitter unit is shown in Fig. 3 and its component layout in Fig. 4. An actual-size, single-side PCB of the receiver unit is shown in Fig. 5 and its component layout in Fig. 6.

After assembling the two units, switch on the power supply of both the units. If the receiver receives the transmitted signal, LED2 will start blinking. The two small digit counters on the right top of the LCD will start changing and the water level (in centimetres) will show on the left-top position. Keep the small spiral antennae in vertical position to have the best radiation. ●



Somnath Bera is an avid user of open source software. Professionally, he is a thermal power expert and works as additional general manager at NTPC Ltd